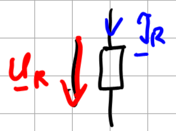
Fortsetzung: $u_R(t) = R \cdot i_R(t)$

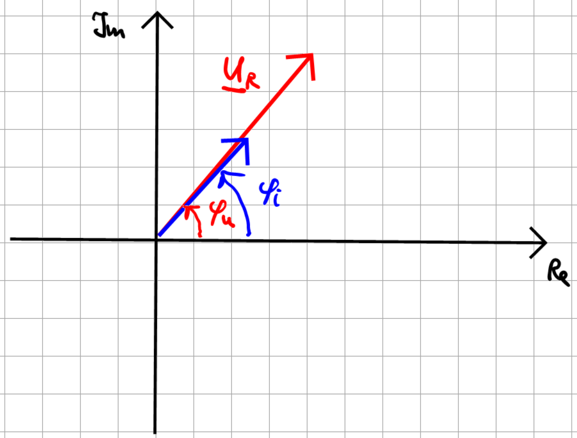
$u_R(t) = \hat{u} \cdot \cos(\omega t + \varphi_u)$ $\qquad$ $i_R(t) = \hat{i} \cdot \cos(\omega t + \varphi_i)$

$\circ\!-\!\bullet$

$\underline{U}_R = \frac{\hat{u}}{\sqrt{2}} e^{j\varphi_u}$ $\qquad$ $\underline{I}_R = \frac{\hat{i}}{\sqrt{2}} e^{j\varphi_i}$

$\varphi_u = \varphi_i = \varphi$





$u_R(t) = R \cdot i_R(t)$

$\hat{u} \cdot \cos(\omega t + \varphi_u) = R \cdot \hat{i} \cdot \cos(\omega t + \varphi_i)$

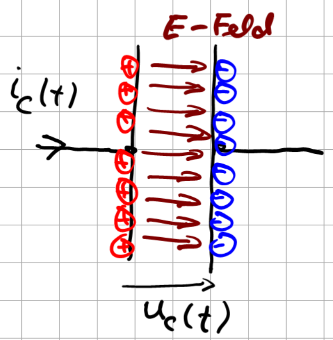
$\circ\!-\!\bullet$ $\qquad$ $\bullet\!-\!\circ$

$\underbrace{\frac{\hat{u}}{\sqrt{2}}} \cdot e^{j\varphi_u} = R \cdot \frac{\hat{i}}{\sqrt{2}} e^{j\varphi_i}$ $\qquad$ $\varphi_u = \varphi_i$

$\underbrace{U \cdot e^{j\varphi_u}} = R \cdot \underbrace{I \cdot e^{j\varphi_i}}$

$$\boxed{\underline{U} = R \cdot \underline{I}}$$ (gilt für Sinusfkt.)

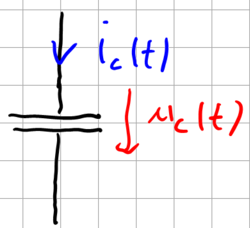
Am <u>Kondensator</u> gilt im Zeitbereich



Im Kondensator gespeicherte Energie

$$W_c = \frac{1}{2} C \cdot U^2$$

$$C = \frac{Q}{U} \Rightarrow U = \frac{1}{C} \cdot \underbrace{Q}_{I \cdot t}$$



$$\boxed{u_c(t) = \frac{1}{C} \int i_c(t)\,dt + U_0}$$ gilt immer!

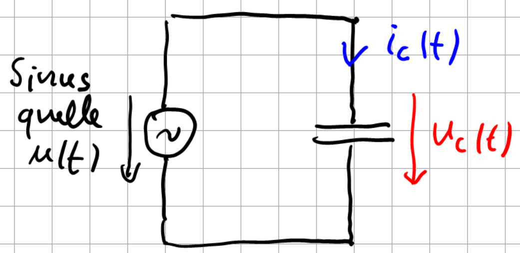
„CDU"

$$\boxed{i_c(t) = C \, \frac{d\,u_c(t)}{dt}}$$ gilt immer!

$$[C] = \frac{As}{V} = \text{Farad} = F$$

$$\Rightarrow \int d\,u_c(t) = \int \frac{1}{C} \, i_c(t)\,dt$$

$$u_c(t) = \frac{1}{C} \int i_c(t)\,dt + \text{const}$$



Sei: $\quad u(t) = u_c(t) = \hat{u} \cdot \cos(\omega t + \varphi_u)$

$$\Rightarrow i_c(t) = C \cdot \frac{d\,u_c(t)}{dt} = C \cdot \frac{d}{dt} \left(\hat{u} \cdot \cos(\omega t + \varphi_u) \right)$$

$$= C \cdot \hat{u} \cdot \left(-\sin(\omega t + \varphi_u) \cdot \omega \right)$$

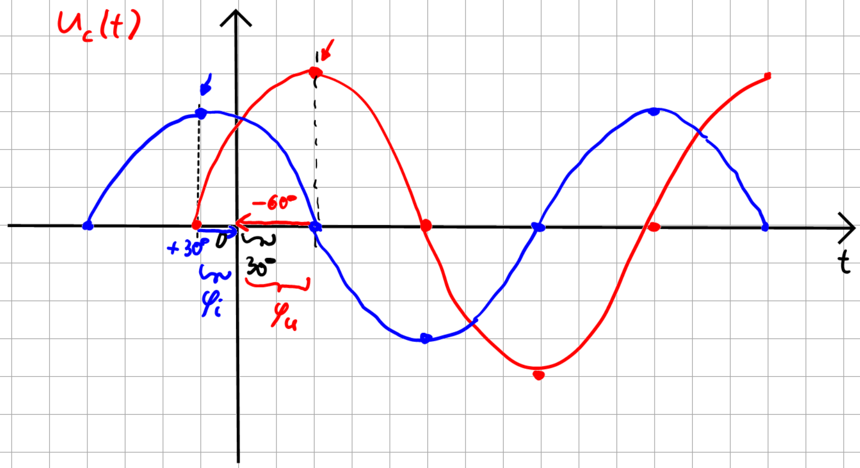
$$= -\omega C \cdot \hat{u} \, \sin(\omega t + \varphi_u)$$

$$\sin(x + 90°) = \cos(x)$$

$$\sin(x) = \cos(x - 90°)$$

$$= -\omega \cdot C \cdot \hat{u} \, \cos(\omega t + \varphi_u - 90°)$$

$$i_c(t) = \underbrace{+\omega C \hat{u}}_{\hat{\imath}} \cos(\omega t + \varphi_u \underbrace{+ 90°}_{\text{Voreilung}})$$



„Am Kondensator eilt der Strom vor."

$$\varphi = \varphi_u - \varphi_i = -60° - 30° = -90° = -\frac{\pi}{2}$$

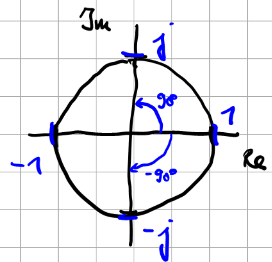
$$i_c(t) = \hat{\imath} \cdot \cos(\omega t + \varphi_i) = \omega C \hat{u} \cos(\omega t + \varphi_u + 90°)$$

$$\frac{\hat{\imath}}{\sqrt{2}} \cdot e^{j\varphi_i} = \frac{\omega C \hat{u}}{\sqrt{2}} e^{j(\varphi_u + 90°)}$$

$$\underbrace{\underline{I} \cdot e^{j\varphi_i}} = \omega C \underbrace{\underline{U} \, e^{j\varphi_u}} \cdot \underbrace{e^{j90°}}_{j}$$

$$\underline{I} = \omega C \cdot \underline{U} \cdot j$$

$$\underline{I} = j \omega C \underline{U}$$



$$\Rightarrow \boxed{\underline{U}_c = \underbrace{\frac{1}{j\omega C}}_{\underline{Z}_c} \underline{I}_c}$$ (gilt für Sinus)

„Impedanz"

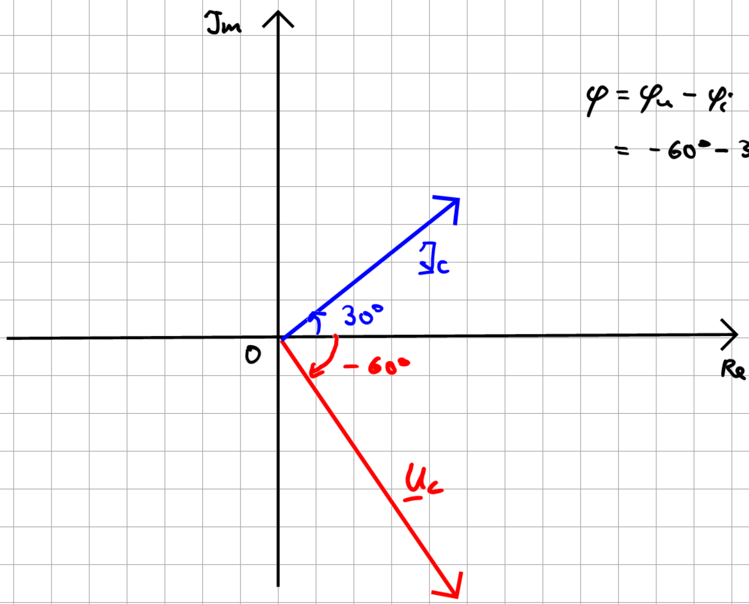
zur Erinnerung: an –▭–

$$\underline{U}_R = R \cdot \underline{I}_R$$

$$\underline{Z}_c = \frac{1}{j\omega C}$$

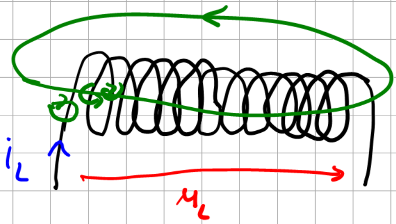
$$|\underline{Z}_c| = \left|\frac{1}{j\omega C}\right| = \frac{|1|}{|j\omega C|} = \frac{1}{|j| \cdot \omega C} = \frac{1}{\omega C}$$

$$\left|\frac{a}{b}\right| = \frac{|a|}{|b|}$$



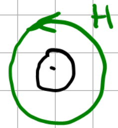
$$\varphi = \varphi_u - \varphi_i$$
$$= -60° - 30° = -90°$$

An der Spule gilt



Wicklung

Magnetischer Energiespeicher



H magnet. Feldstärke

$$B = \mu \cdot H = \mu_0 \cdot \mu_r \cdot H$$

↑ Flussdichte

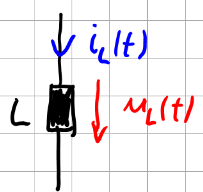
Energie der Spule:

$$\boxed{W_L = \frac{1}{2} L \cdot i^2}$$

$$L = \frac{\Psi}{i}$$ Induktivität

$$[L] = \frac{Vs}{A} = \text{Henry} = H$$

Schaltsymbol



$$\boxed{u_L(t) = L \cdot \frac{d\, i_L(t)}{dt}}$$ gilt immer!

$$u_L = L \cdot \dot{i}_L$$

$$\int d i_L(t) = \int \frac{1}{L} u_L(t)\, dt$$

$$\Rightarrow \boxed{i_L(t) = \frac{1}{L} \int u_L(t)\, dt + I_0}$$

$$u_L(t) = L \cdot \frac{d\, i_L(t)}{dt}$$

$$\text{Sei: } \underbrace{i_L(t) = \hat{\imath} \cdot \cos(\omega t + \varphi_i)}$$

$$\Rightarrow \quad u_L(t) = L \cdot \frac{d}{dt}\left(\hat{\imath} \cdot \cos(\omega t + \varphi_i)\right)$$

$$= L \cdot \hat{\imath} \; (-\sin(\omega t + \varphi_i) \cdot \omega)$$

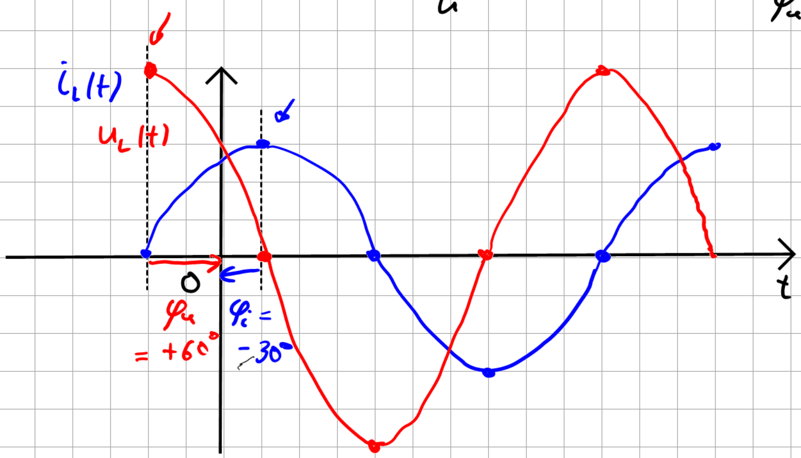
$$= \omega \cdot L \cdot \hat{\imath} \cdot (-\sin(\omega t + \varphi_i))$$

$$= -\omega L \cdot \hat{\imath} \; \cos(\omega t + \varphi_i - 90°)$$

$$= \underbrace{\omega L \hat{\imath}}_{\hat{u}} \; \cos(\omega t + \underbrace{\varphi_i + 90°}_{\varphi_u})$$

$-\sin x$

$$u_L(t) = \hat{u} \cdot \cos(\omega t + \varphi_u) = \underbrace{\omega L \hat{\imath}}_{\hat{u}} \cdot \cos(\omega t + \underbrace{\varphi_i + 90°}_{\varphi_u})$$



$$\varphi = \varphi_u - \varphi_i = 60° - (-30°) = +90°$$

„An Induktivitäten die Ströme sich verspäten"

$$u_L(t) = \hat{u} \; \cos(\omega t + \varphi_u) = \omega L \; \hat{\imath} \; \cos(\omega t + \varphi_i + 90°)$$

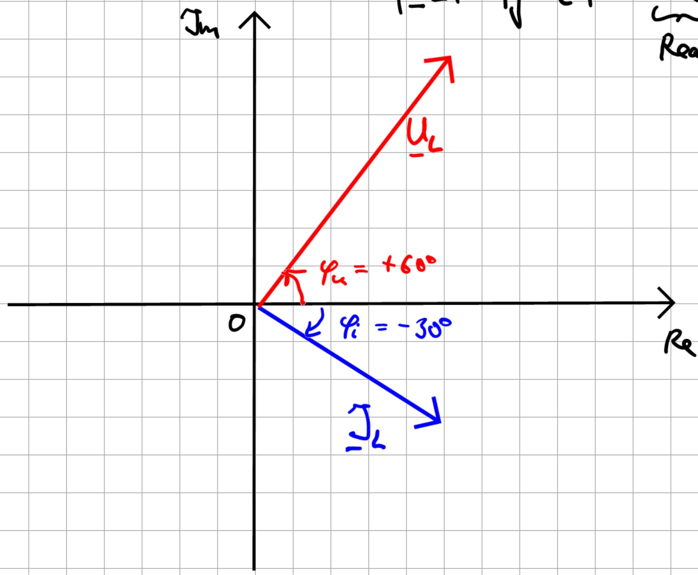
$$\underbrace{U \cdot e^{j\varphi_u}}_{\underline{U}} = \omega L \; I \; e^{j(\varphi_i + 90°)}$$

$$= \omega L \cdot \underbrace{I \cdot e^{j\varphi_i}}_{\underline{I}} \cdot \underbrace{e^{j90°}}_{j}$$

$$\boxed{\underline{U}_L = \underbrace{j\omega L}_{\underline{Z}_L} \cdot \underline{I}_L}$$

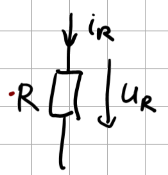
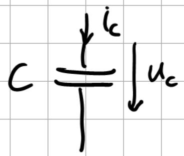
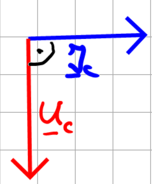
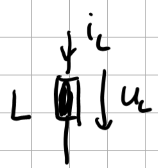
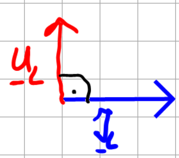
$\underline{Z}_L$ Impedanz der Spule

$$|\underline{Z}_L| = |j\omega L| = \underbrace{\omega L}_{\text{Reaktanz}}$$



$$\underline{\underline{\varphi}} = \varphi_u - \varphi_i = 60° - (-30°) = \underline{\underline{+90°}}$$

Wiederholung:

	Zeitbereich	Frequenzbereich	
	$u_R(t) = R \cdot i_R(t)$	$\boxed{\underline{U}_R = R \cdot \underline{I}_R}$	
	$i_C(t) = C \dfrac{du_C(t)}{dt}$	$\boxed{\underline{U}_C = \dfrac{1}{j\omega C} \cdot \underline{I}_C}$	
	$u_L = L \cdot \dfrac{di_L(t)}{dt}$	$\boxed{\underline{U}_L = j\omega L \cdot \underline{I}_L}$	

$$\frac{1}{j} = -j$$

$$\frac{1}{j} = \frac{1}{e^{j\frac{\pi}{2}}} = e^{-j\frac{\pi}{2}} = -j$$